Mutual Fund Analytics

Internship Capstone Project Report

Submitted By:

Suhana Begum

College:

Malla Reddy Engineering College for Women

Organization:

BlueStock

Submission Date:

12 June 2026

Executive Summary

This project was completed as part of my internship at Bluestock. The main objective of the project was to collect, clean, analyze, and visualize mutual fund data to gain meaningful insights about the mutual fund industry.

The project involved working with multiple datasets related to mutual funds, including scheme information, NAV data, AUM data, and industry statistics. An ETL pipeline was developed using Python to process the data and store it in a SQLite database. Exploratory Data Analysis (EDA) was performed to understand trends and patterns in the data.

Performance metrics such as Sharpe Ratio, Alpha, Beta, and Rolling Sharpe Ratio were calculated to evaluate fund performance. A recommendation system was also created to suggest mutual funds based on risk appetite. Finally, a Power BI dashboard was developed to present the insights in an interactive and user-friendly manner.

This project helped me gain practical experience in data engineering, data analysis, database management, visualization, and financial analytics.

1. Introduction

Mutual funds have become one of the most popular investment options for investors. They allow individuals to invest in a diversified portfolio managed by professional fund managers.

The mutual fund industry generates large amounts of data related to fund performance, assets under management, investor participation, and market trends. Analyzing this data can help investors and financial institutions make better decisions.

The purpose of this project was to build an end-to-end analytics solution that collects, processes, analyzes, and visualizes mutual fund data.

2. Project Objectives

The main objectives of this project were:

1. Collect mutual fund related datasets.
2. Clean and preprocess the data.
3. Store processed data in a SQLite database.
4. Perform exploratory data analysis.
5. Calculate fund performance metrics.
6. Build a mutual fund recommendation system.
7. Create an interactive Power BI dashboard.
8. Generate insights and recommendations.

3. Data Sources

The project used multiple CSV files containing mutual fund related information.

The datasets included:

- Mutual fund scheme details
- Historical NAV data
- Assets Under Management (AUM)
- SIP inflow statistics
- Industry-level mutual fund data
- Fund category information
- Performance related data

A total of multiple CSV files were collected and processed as part of the ETL pipeline.

4. Tools and Technologies Used

The following tools were used during the project:

Programming Language

Python

Libraries

- Pandas
- NumPy
- Matplotlib
- SQLite3

Database

SQLite

Visualization Tool

Power BI

Development Environment

Visual Studio Code

Version Control

Git and GitHub

These tools helped in performing data cleaning, analysis, database management, and dashboard creation.

5. ETL Process

ETL stands for Extract, Transform, and Load.

The ETL pipeline was created to automate data processing.

Extract

The raw data was collected from CSV files.

Transform

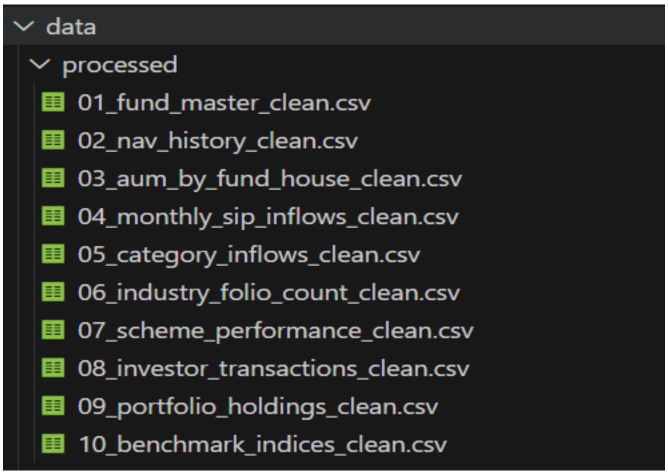
The data was cleaned by:

- Handling missing values
- Removing duplicates
- Standardizing column names
- Correcting data types

Load

The cleaned data was loaded into a SQLite database for further analysis.

This process ensured that the data was accurate and ready for analysis.



6. Database Design

SQLite was used as the database system for storing processed data.

The database helped organize the datasets and made querying easier.

Benefits of using SQLite:

- Lightweight
- Easy to manage
- No server setup required
- Suitable for analytics projects

The database stored multiple tables corresponding to the processed datasets.

7. Exploratory Data Analysis (EDA)

EDA was performed to understand the data before performing advanced analysis.

Several visualizations were created to identify patterns and trends.

The analysis focused on:

- Fund category distribution
- AUM distribution
- Industry trends
- Top performing funds
- Risk and return relationship

EDA helped in identifying important insights from the data.

8. EDA Findings

Some important observations from the analysis were:

Fund Categories

Equity funds represented a significant portion of the dataset.

Assets Under Management

A small number of schemes accounted for a large percentage of total AUM.

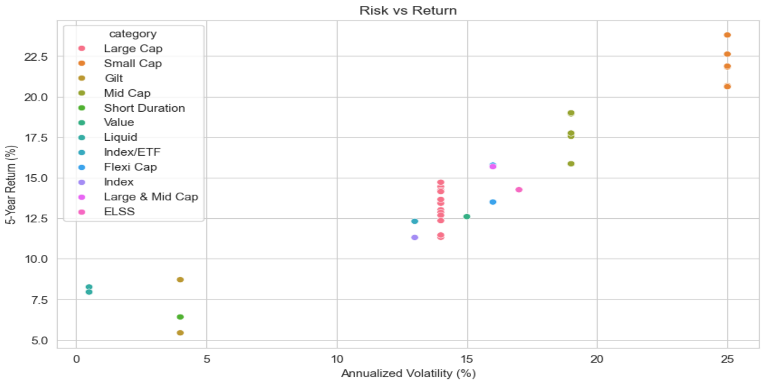
Industry Trends

The mutual fund industry showed steady growth in investor participation.

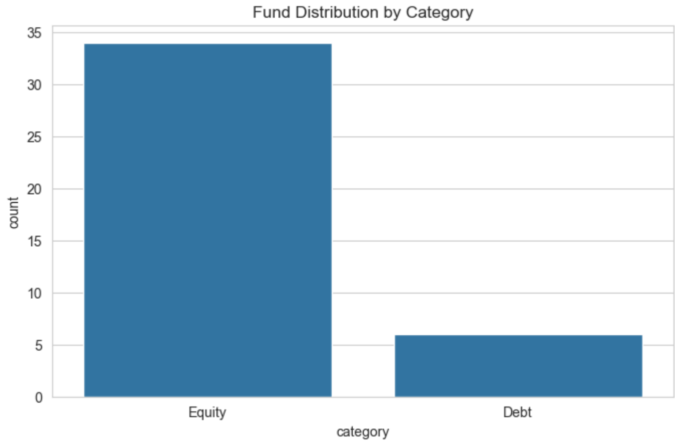
Risk and Return

Funds with higher returns generally showed higher risk levels.

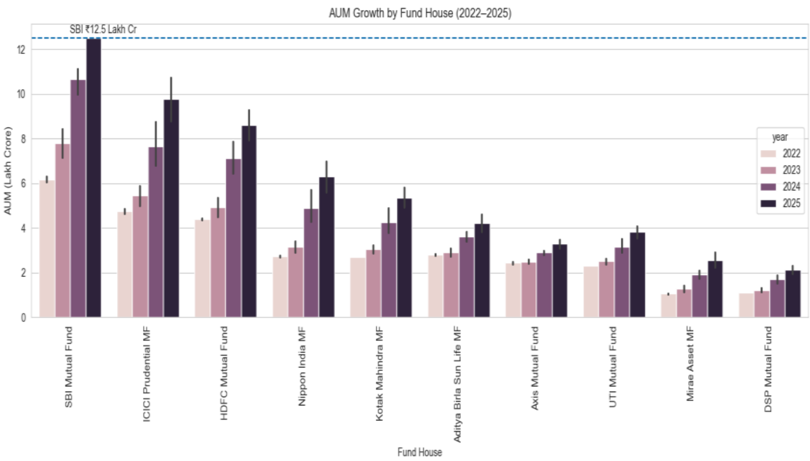
Risk VS Return Analysis:



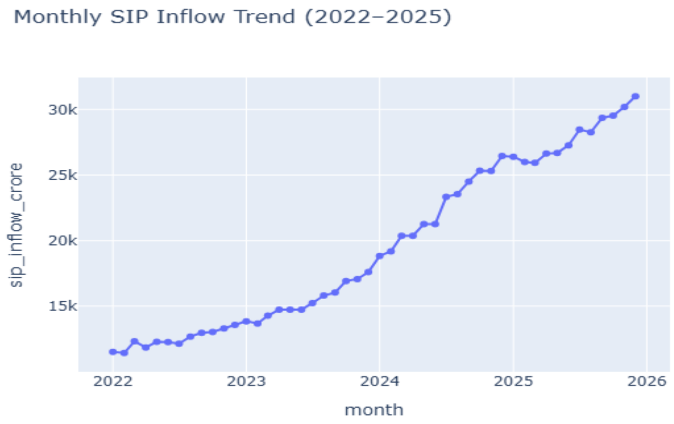
Category Analysis:



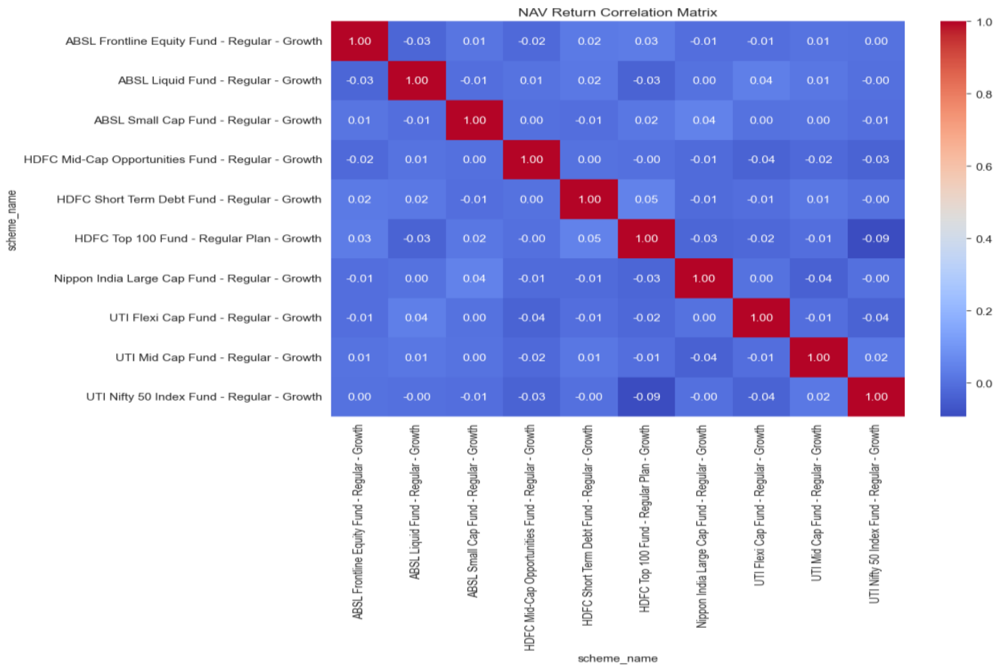
AUM Analysis:



Monthly SIP Trend:

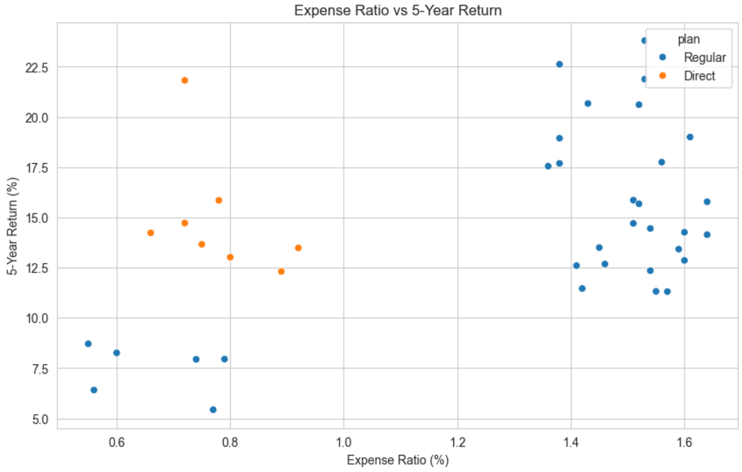


Correlation Analysis:

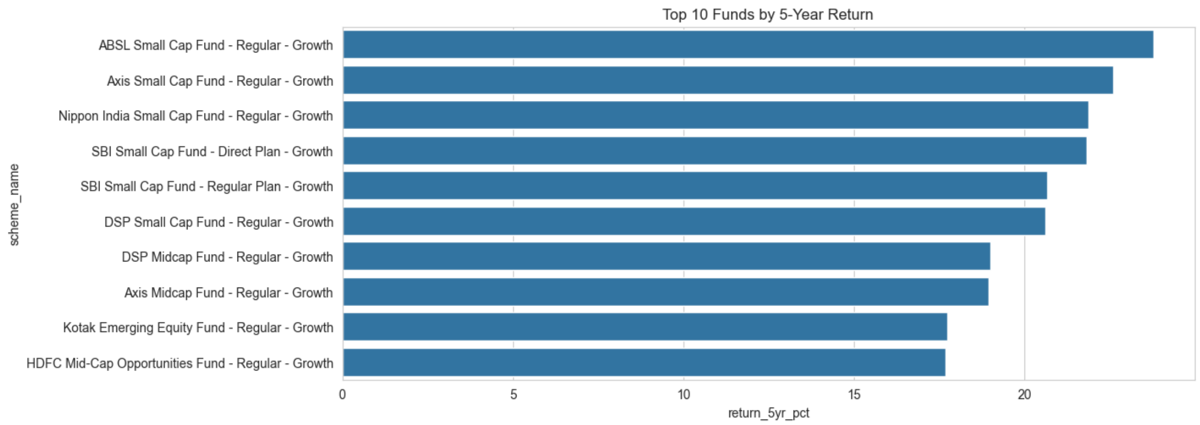


Folio Growth:

Expenses VS Returns:



Top 10 Returns:



9. Performance Analytics

To evaluate fund performance, several financial metrics were calculated.

These metrics help investors compare different funds.

The metrics used were:

- Sharpe Ratio
- Alpha
- Beta
- Rolling Sharpe Ratio

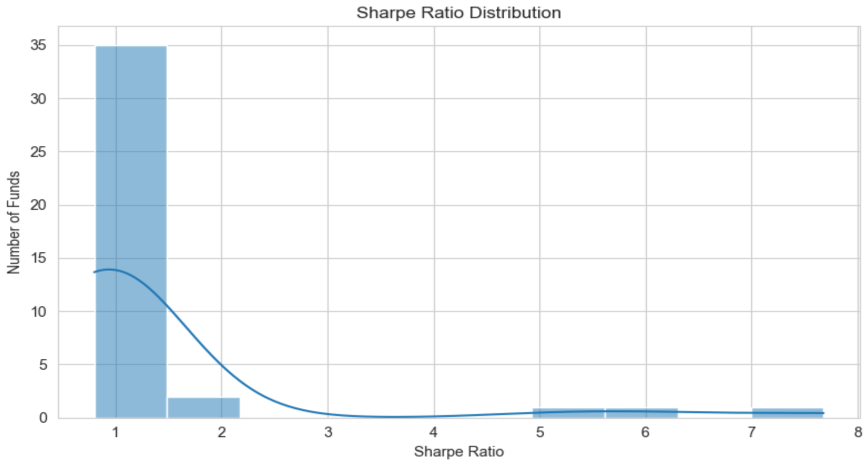
10. Sharpe Ratio Analysis

Sharpe Ratio measures the return generated for each unit of risk taken.

Higher Sharpe Ratios generally indicate better risk-adjusted performance.

The analysis showed that some funds consistently performed better than others after adjusting for risk.

Sharpe Ratio Chart:



11. Alpha and Beta Analysis

Alpha measures the excess return generated compared to the benchmark.

Beta measures the sensitivity of a fund relative to market movements.

Key observations:

- Positive Alpha indicates better-than-expected performance.
- Beta greater than 1 indicates higher volatility.
- Beta less than 1 indicates lower volatility.

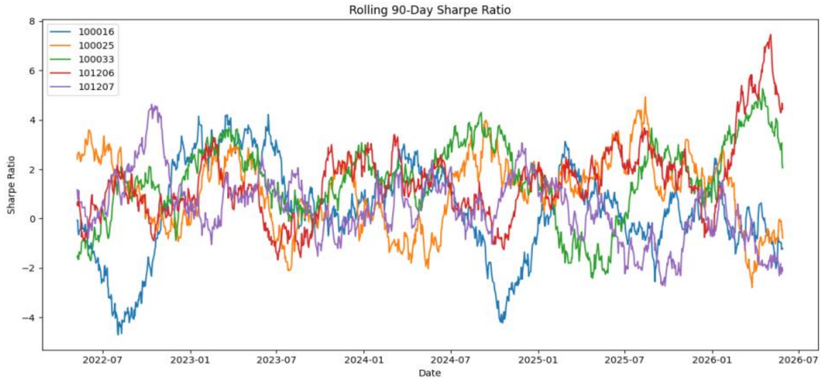
12. Rolling Sharpe Analysis

Rolling Sharpe Ratio was used to study how risk-adjusted performance changed over time.

This metric provides a more dynamic view compared to a single Sharpe Ratio value.

The analysis showed that performance consistency varied across funds.

Rolling Sharpe Chart:



13. Mutual Fund Recommendation System

A recommendation system was developed using Python.

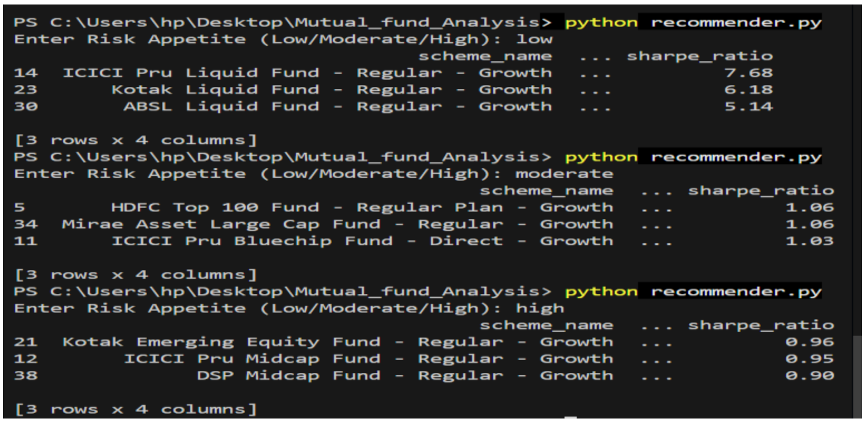
The system suggests suitable funds based on the user's risk appetite.

Supported risk categories:

- Low Risk
- Moderate Risk
- High Risk

The recommendation engine helps users identify funds that match their investment preferences.

Recommender Output:



```
PS C:\Users\hp\Desktop\Mutual_fund_Analysis> python recommender.py
Enter Risk Appetite (Low/Moderate/High): low
                                 scheme_name  ... sharpe_ratio
14  ICICI Pru Liquid Fund - Regular - Growth  ...         7.68
23      Kotak Liquid Fund - Regular - Growth  ...         6.18
30       ABSL Liquid Fund - Regular - Growth  ...         5.14

[3 rows x 4 columns]
PS C:\Users\hp\Desktop\Mutual_fund_Analysis> python recommender.py
Enter Risk Appetite (Low/Moderate/High): moderate
                                      scheme_name  ... sharpe_ratio
5       HDFC Top 100 Fund - Regular Plan - Growth  ...         1.06
34  Mirae Asset Large Cap Fund - Regular - Growth  ...         1.06
11      ICICI Pru Bluechip Fund - Direct - Growth  ...         1.03

[3 rows x 4 columns]
PS C:\Users\hp\Desktop\Mutual_fund_Analysis> python recommender.py
Enter Risk Appetite (Low/Moderate/High): high
                                      scheme_name  ... sharpe_ratio
21  Kotak Emerging Equity Fund - Regular - Growth  ...         0.96
12       ICICI Pru Midcap Fund - Regular - Growth  ...         0.95
38             DSP Midcap Fund - Regular - Growth  ...         0.90

[3 rows x 4 columns]
```

14. Power BI Dashboard

A Power BI dashboard was developed to present the analysis visually.

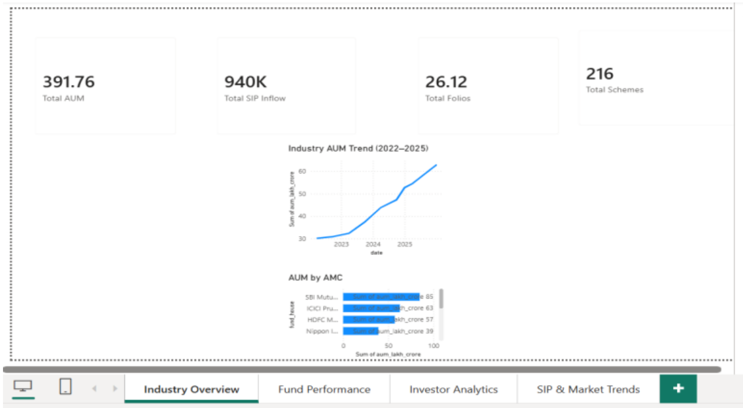
The dashboard contains four pages:

1. Industry Overview
2. Category Analysis
3. Fund Performance
4. Recommendation Insights

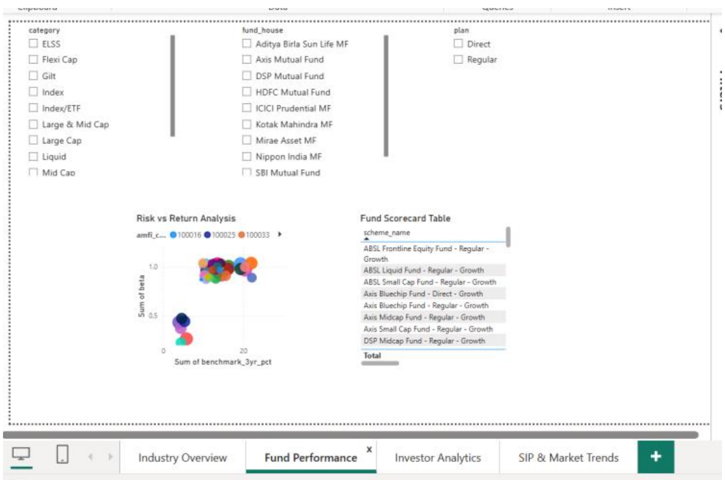
The dashboard includes:

- KPI Cards
- Charts
- Filters
- Slicers
- Interactive Visualizations

Page 1:

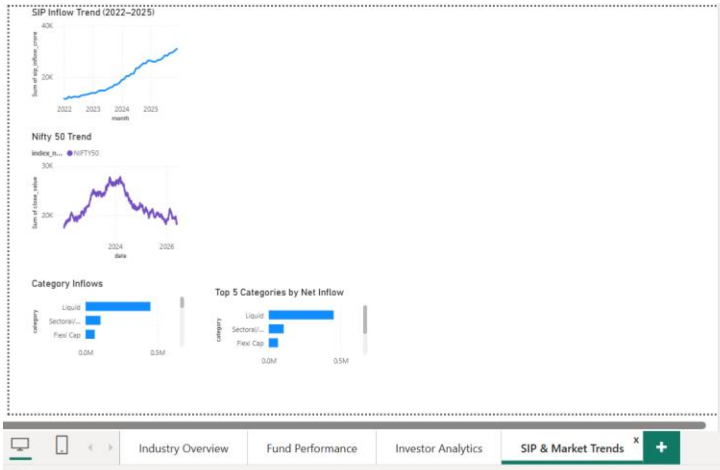


Page 2:



Page 3:

Page 4:



15. Key Findings

The major findings from this project are:

- Equity funds dominate the industry.
- A few funds contribute significantly to total AUM.
- Risk-adjusted performance varies across funds.
- Some funds consistently outperform benchmarks.
- Investor participation in mutual funds continues to increase.

These insights can help investors make informed investment decisions.

16. Challenges Faced

During the project, several challenges were encountered:

- Handling missing values in datasets.
- Managing data from multiple files.
- Ensuring consistency across datasets.
- Creating meaningful performance metrics.
- Designing an interactive dashboard.

These challenges helped improve problem-solving skills and technical knowledge.

17. Limitations

The project has some limitations:

- Analysis is based on available historical data.
- Market conditions may change in the future.
- Recommendations are based on selected metrics.
- Additional external data sources could improve accuracy.

18. Future Improvements

The project can be enhanced in several ways:

- Integration of live market data.
- Machine learning based recommendations.
- Cloud database implementation.
- Web application deployment.
- Real-time dashboard updates.

19. Conclusion

The Mutual Fund Analytics project successfully achieved its objectives of collecting, processing, analyzing, and visualizing mutual fund data.

The project provided valuable insights into fund performance, industry trends, and investment behavior. The combination of Python, SQLite, and Power BI helped create a complete analytics solution.

Through this internship project, I gained practical experience in data analytics, database management, visualization, and financial data analysis. This project also improved my understanding of real-world data workflows and business intelligence tools.

References

1. AMFI Industry Reports
2. Power BI Documentation
3. Python Documentation
4. Pandas Documentation
5. SQLite Documentation
6. Mutual Fund Industry Publications

THANK YOU